Dadmehr Daniel Ghasemfar

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Summary

Duke mechanical engineer with a background in robotics, applied math, and aerospace. Experienced in AI integration and advanced prototyping. Skilled in CAD, FEA/CFD, and numerical methods.

Technical Skills

Languages: Java, Python, C++, MATLAB, Maple, TensorFlow, HTML/CSS/JavaScript
Software: CAD [SolidWorks, Fusion, OnShape], FEA and CFD [Ansys, SolidWorks, In-House], PCB [Eagle, KiCad]
Equipment: 3D Printing [SDF, SLS, Metal], Machining [Mill, CNC, Lathe, Woodwork], Lasercutting, Soldering
Theoretical: solving PDE's and ODE's, Numerical Analysis, Accurate GPU Simulations, PID's and Advanced Controls
Soft Skills: 6-month Project Management, Clear Communication, Professional Documentation

Education

Duke University — Durham, NC
Bachelor of Science in Mechanical Engineering with Minors in Math and Psychology — Aug 2020 – Dec 2024

- Total GPA: 3.48 → Engineering GPA: 3.56 → Math GPA: 3.5 → Psych. GPA: 3.0
- Classes: intro/intermediate robotics, intermediate dynamics, chaos and perturbation theory, control theory, heat and mass transfer, incompressible fluid dynamics, algorithms and data structures, ODE/PDE's, numerical analysis, intermediate solids and statics, aircraft performance and design
- Graduated as a Pratt Fellow for recognition of a 2+ year research project

Experience

3DA Solutions [thinktank] — Aug 2024 – Present
Co-Founder — Alexandria, VA

- Worked with Dr. Siavash Gholamhosseinpour to design, protoype, and build a novel FDS, Insulation Resistance and HiPOT Integrated Tester. It costs and weighs $\frac{1}{3}$ of the closest competition in a $700M market
- Consulted with a handful of IP-law firms and negotiated deals with several potential private investors. Created the plan for non-provisional patent acquisition and legal entity creation
- Helped edit a 120-page technical datasheet and a 70-page professional plan

Duke General Robotics Lab — Aug 2022 – May 2024
Individual Research — Durham, NC

- Worked with Dr. Boyuan Chen to build and develop a versatile and accurate particle parallel-processed wave propagation simulation. This could emulate any material and lattice, 3D shape, and input forces. Used to generate artificial data to train a custom A.I. that can "see with sound"
- Created a novel Chladni plate comparison test to connect simulated data to real-world vibration data. See website portfolio for the white paper and results
- Research approved by Duke Engineering Board for *Graduation with Distinction award*

FPrin LLC — Oct 2020 – Feb 2021
Lab Technician — Mountain View, CA

- Designed, built, and operated case-specific test devices to effectively characterize a recently patented German compressed gas-powered insulin pen. After that, stress tested thousands of components and maintained the shop/lab
- Designed and manufactured a fully functional self-balancing RC PID controlled robot named Balanciaga for as an internal project

Extracurriculars

Clubs & Activities & Hobbies

- High School XC Mens Captain
- 3x XC States Championships
- California Mayor Medal for A.I. forest fires study
- 35 ACT Score, 800 Math and Physics SATs
- 6+ years Calculus/Trig Tutor
- FRC Robotics Quarterfinals
- Duke 24-hour Design Sprint Winner
- 2x Synopsys Science Fair Finalist [CS]
- Los Altos Hackathon III organization